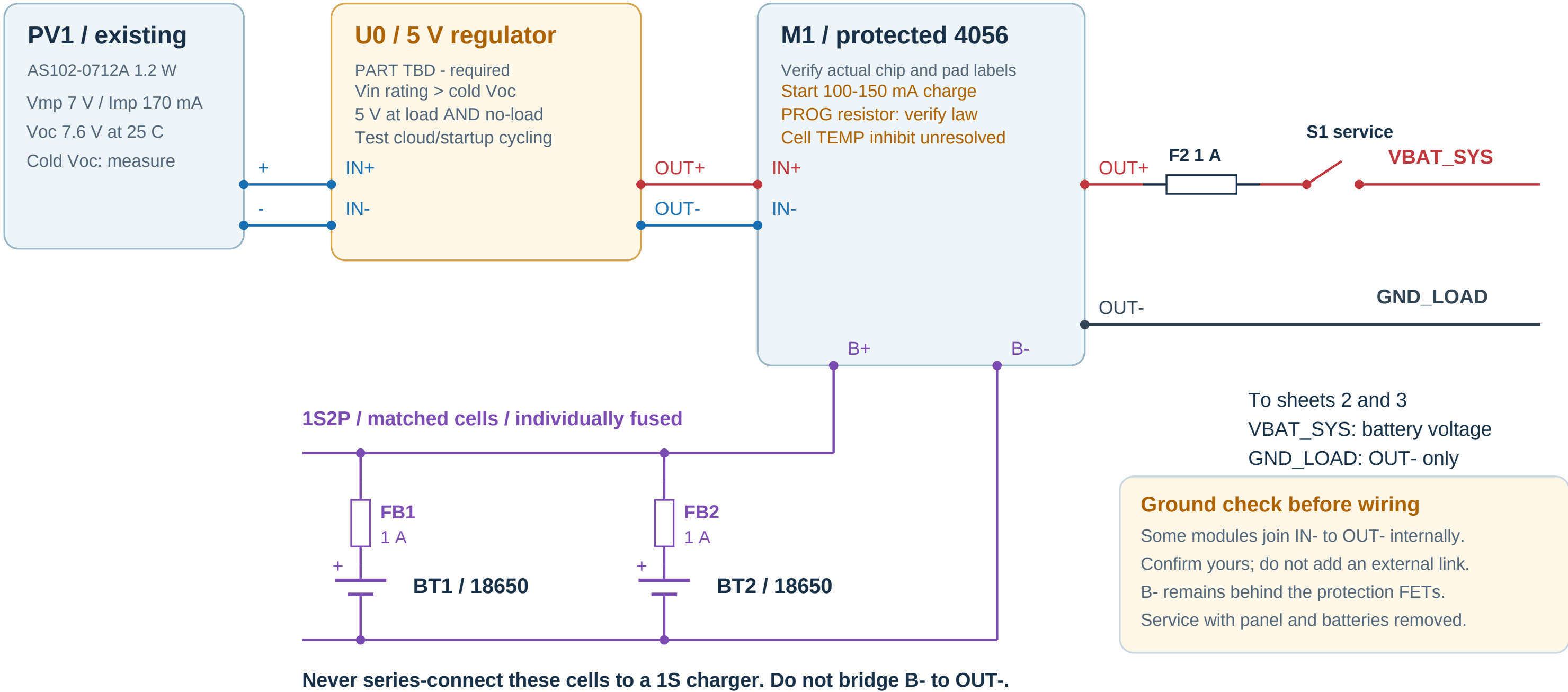


Power, charging & battery

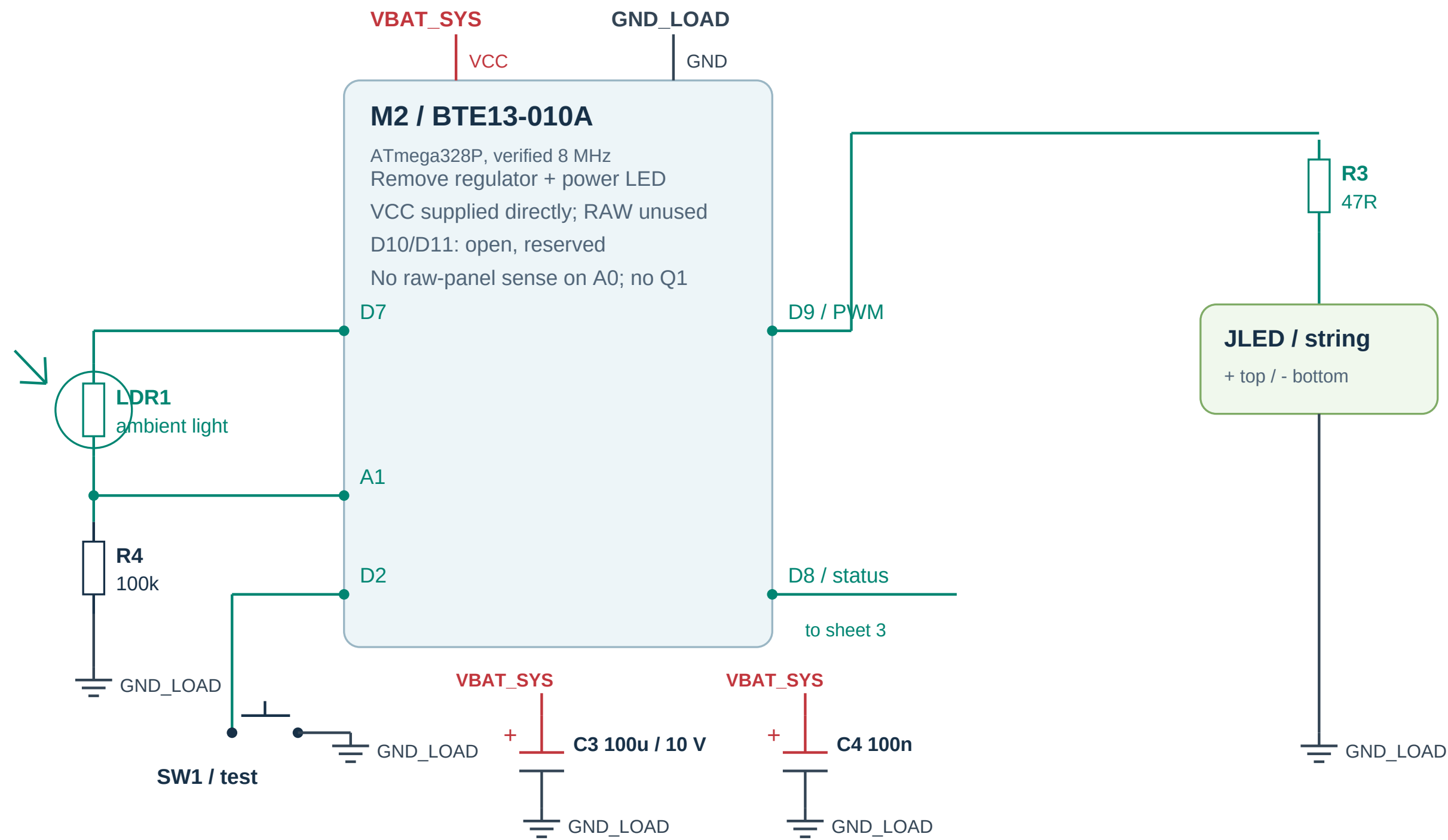
Terminal-level schematic. Amber blocks need part selection and bench qualification. All loads connect to protected outputs.



No pair of series diodes is accepted as overvoltage protection. Unattended charging awaits cell-temperature protection.

Light controller & LED driver

Reused 16 MHz Pro Mini converted to internal 8 MHz. Switched LDR; dusk to dawn at 100% command, with R3 setting the current.



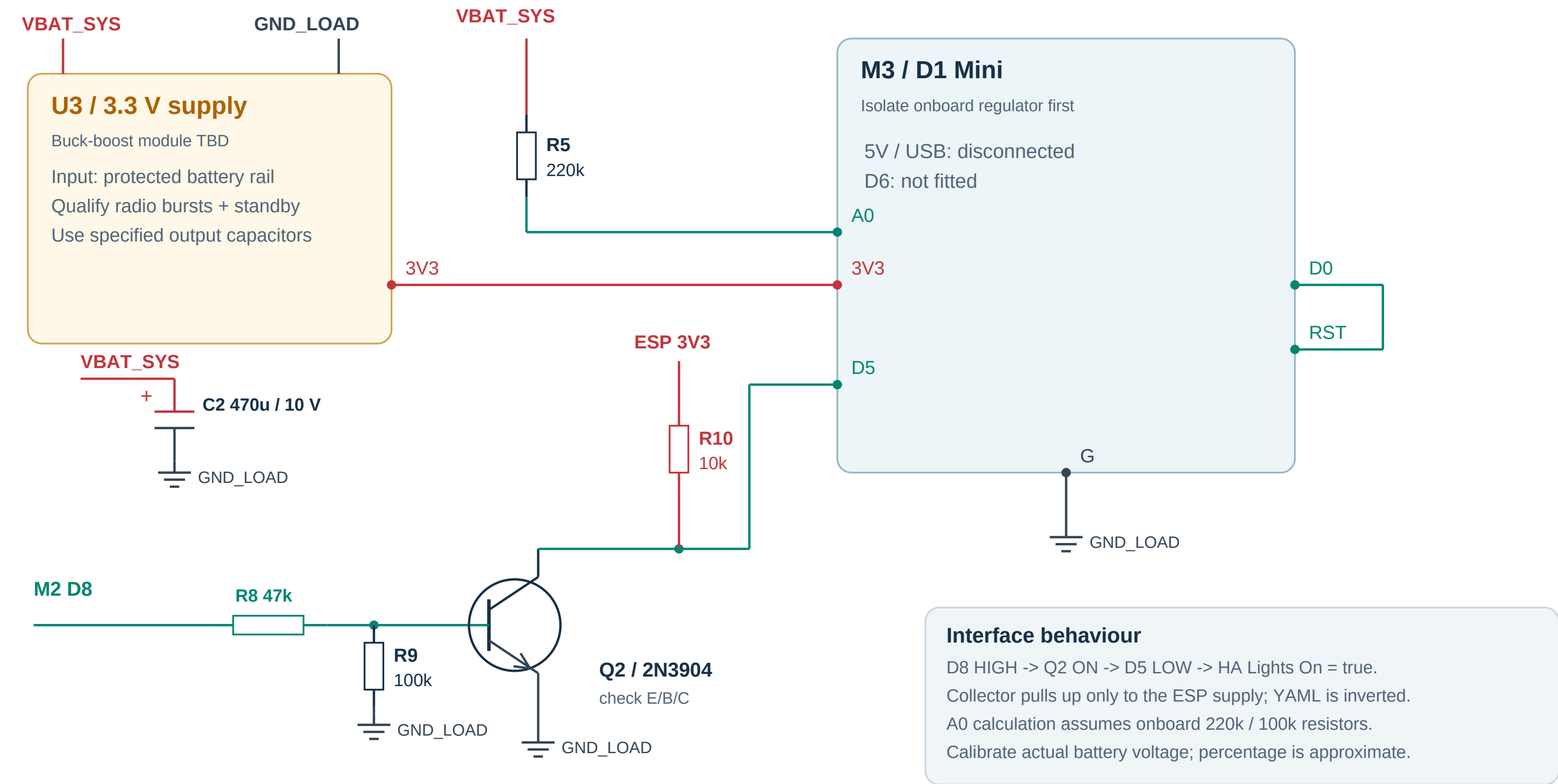
No Q1, R6, R7 or F1

Measured string: 2.56 V drop
 $I = (VBAT - 2.56) / (R3 + 30R)$
21 / 15 / 11 mA at
4.20 / 3.70 / 3.40 V
Fit 56R to stay under 20 mA
Pin max 40 mA absolute
Fit R3 at the board end

All ground symbols on this sheet mean OUT-. R3 sets the peak current; duty changes the average only. Verify current at 4.20 V and 100% duty.

Optional Wi-Fi telemetry

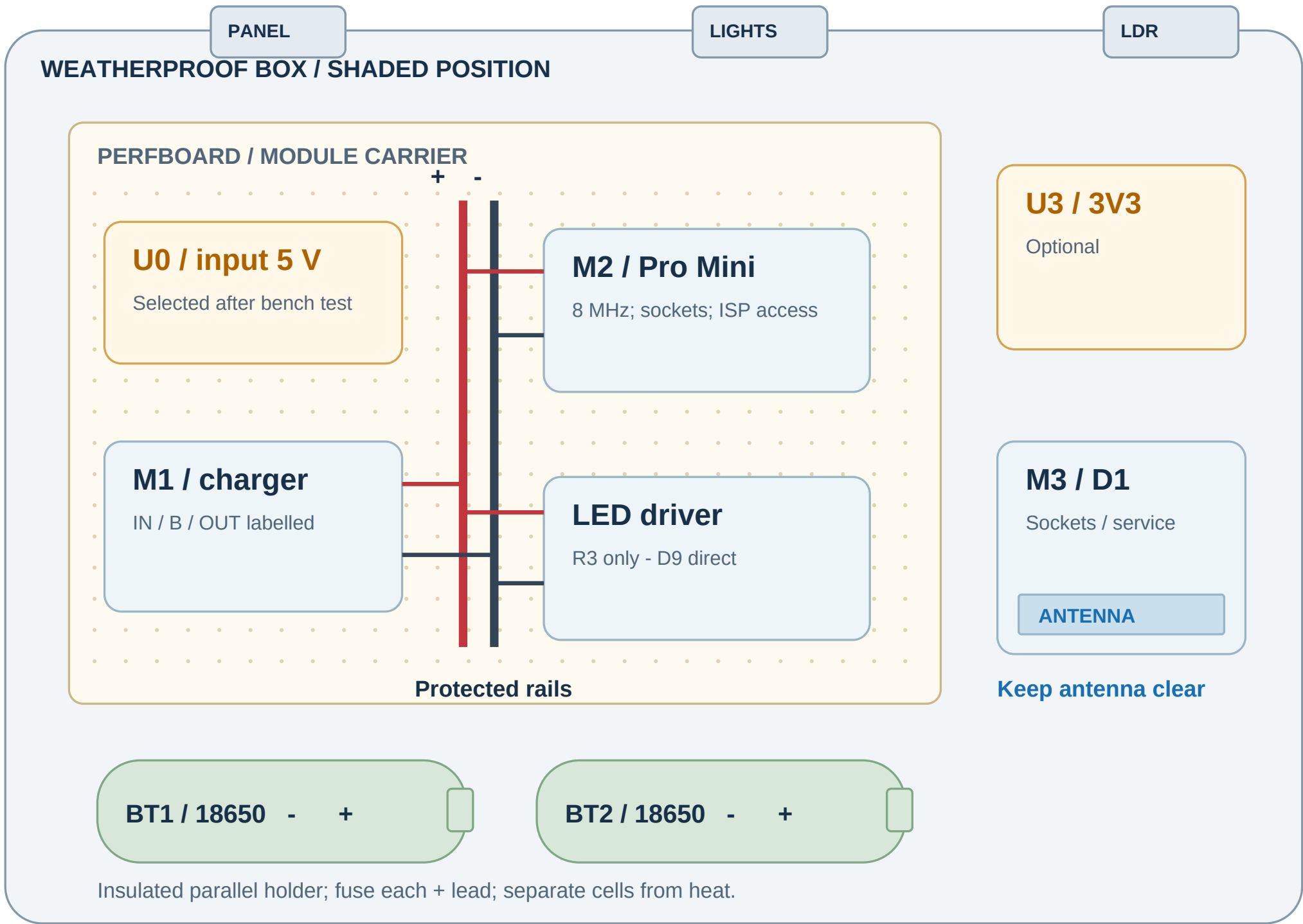
Add after the lights work. D1 Mini reports battery and lights only. No CHRG connection in this revision.



Disconnect external power AND signal leads before USB/programmer servicing. Never supply the ESP 3V3 pin from the cell directly.

What to assemble inside the enclosure

Top-down placement concept, not a drilling template. Use terminal labels and the wire schedule for electrical connections.



Build in this order

- 1 Qualify panel regulator + charger.
- 2 Convert and test the Pro Mini.
- 3 Add R3 (47R) and light string - no MOSFET.
- 4 Add LDR; check dusk/dawn + LVC.
- 5 Assemble the protected, fused pack.
- 6 Add optional telemetry and measure.

Label every connector

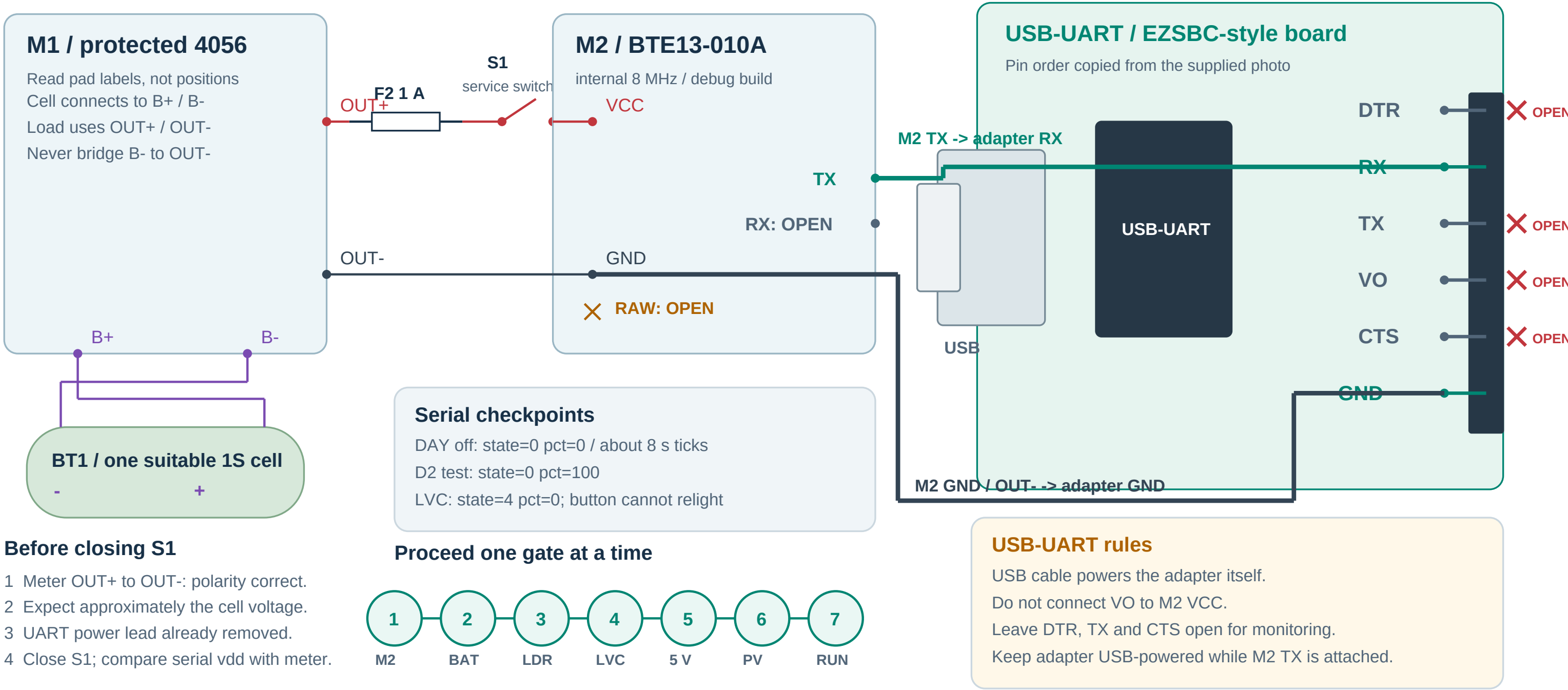
RED: protected positive / VBAT_SYS
BLACK: load return / OUT-
BLUE: panel input pair
PURPLE: battery-only connections
GREEN: logic / sensor signals

Before closing the lid

Strain relief, polarity, cell insulation.
No exposed underside copper.
Charger/cell temperature tests passed.
Log charge, load and actual brightness.

Staged power-up & serial diagnostics

Battery powers the controller. The USB-UART listens only: connect RX and GND; leave its power and control pins open.



SERIAL IS NOT A CHARGE METER: verify panel input, regulated 5 V, battery voltage and charge current with instruments.